# Ping-Jung Wu

## Education

**M.Sc. in Aerospace** **10/2024 – present**
Technical University of Munich (TUM), Munich, Germany
**Current GPA:** 1.5
**Key Focus:** UAV Control, Trajectory Optimization, Autonomous Navigation
**Selected Projects:**

- **Drone SLAM Navigation**: Implemented visual mapping and navigation architecture for drone tunnel search
- **Rocket Altitude Control**: Designed simulations in a gas-dynamic system by implementing an MPC controller for altitude stabilization
- **Physics-Informed AI**: Implemented deep learning models for solving PDEs in solid mechanics, heat transfer, and traffic-flow dynamics

**Related Courses:** Aircraft Trajectory Optimization, Optimal Control and Decision Making, Nonlinear Flight Control, Autonomous System

**B.Sc. in Mechanical Engineering** **09/2020 – 01/2024**
National Taiwan University (NTU), Taipei, Taiwan
**GPA:** 3.97/4.30
**Key Focus:** Linear Control, Fluid Mechanics
**Related Courses:** Flight Dynamics, Viscous Flow, Linear Control Systems

## Research Experience

**Master's Thesis**, Technical University of Munich **05/2026 –**

**Coordinated Autonomous Exploration in Forests for Decentralized Multi-UAV Systems**

- Develope a ROS2/C++ decentralized autonomous stack featuring ad-hoc meeting and spatial allocation protocols for multi-agent coordination in dense canopy search and mapping
- Reduction in mission time and a decrease in average path length compared to baseline peer research across varying forest densities in simulation
- Preparing PX4 autopilot integration for real-world swarm flight validation

**Semester Thesis**, Technical University of Munich **10/2025 – 04/2026**

- Validated quadrotor's trajectory optimization methods from the frontline research
- Designed minimum control based optimizers under safe flight corridors
- Developed unified collision-avoidance constraints for a multi-agent system within a centralized architecture

**Bachelor's Thesis,** National Taiwan University **02/2022 – 07/2022**

- Constructed a control architecture by using setpoint tracking for tandem wing tail-sitter
- Implemented a cascaded PID controller for longitudinal control
- Validated aircraft dynamic responses through CFD simulations and longitudinal stability analysis

**Personal Projects** **07/2021 – 07/2023**

- **Attitude and Heading Reference System:** Developed an Arduino-based attitude and heading reference system using an Extended Kalman Filter for sensor fusion.

- **Fixed Wing Aircraft:** Constructed a monoplane by utilizing Thin Airfoil Theory
- **Tandem Wing Tail-sitter:** Built a concept aircraft with the capability for both vertical and horizontal takeoff and landing

## Skills

**Programming:** Python, PyTorch, C / C++, ROS 2, MATLAB, Git
**Languages:** Mandarin (native), English (TOEFL 100 / 120), German (B2, currently improving), Japanese (basic)
**Software:** Ansys Fluent, Autodesk Inventor, AutoCAD, SolidWorks

## Awards

**1st Place — International Micro Air Vehicles (IMAV) Competition,** Strasbourg, France **2026**

- Represented Horyzn Software Team and secured 1st place in the IMAV indoor drone competition.
- Developed a LiDAR-based waypoint navigation system enabling reliable obstacle-avoidance mission execution.
- Built Gazebo simulation and performed sim-to-real verification for autonomous flight.

**Dean's List Award,** National Taiwan University **Spring 2022, Spring 2023**

- Awarded top five percent of students for excellent academic performance

**Masterpiece Award for Bachelor's Thesis,** National Taiwan University **2023**

- Awarded by the College of Engineering for excellent research work

## Club Activities

**Software Developer**, Horyzn, Technical University of Munich **05/2026 – present**

- Integrate and deploy UAV trajectory-planning algorithms within PX4 simulation
- Develop a vision-assisted waypoint navigation system with obstacle detection

**Lecturer**, NTU Aviation Club, National Taiwan University **09/2022 – 12/2023**

- Subjects: Basic Aerodynamics, Model Aircraft Manufacturing